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Chapter 2

Processes



Job Schedules

Use the Job Schedules function to maintain the jobs scheduled to be run. The following jobs will be included in the schedule.

1. Auto Day-Start and Auto Day-End Jobs due to run today with explicit schedule times.
2. Jobs scheduled using the prompt facility with Application Manager. See the Application Manager Product Guide, section Submitting Batch Jobs to the Machine Manager Scheduler.
3. Jobs you have added directly to the schedule using this function.
4. The scheduler will contain Auto Day-Start Jobs without explicit times, that are due to be run today, these will be executed in order of their sequence numbers. These jobs will be included in the scheduler by the Machine Manager System Start Up program. See the Power Down Times section in this chapter.
5. The scheduler will contain Auto Day-End Jobs without explicit times, that are due to be run today, these will be executed in order of their sequence numbers. These jobs will be included in the scheduler by the Machine Manager end day function. See the End-Day Times section.

MM141 Machine Manager System: STUDB30A

Manage Jobs Scheduled

Type option, press Enter
2=Change, 5=Display

Current Date: 5/07/91
Time: 17:22:05
Next active: 21:08:16

Opt	Time	Seq.No.	Description
-	5:00:00	1	Initiate End Day Program
-	12:00:00	2	Run Business Modeller Reports
-	16:00:00	3	Send early close message to all users

Bottom

Change schedule time of selected jobs to ..

F3=Exit F8=Add

Fields:

Opt

Enter 2 to maintain any existing jobs.

Enter 5 to display the details of existing jobs.



Note: If you maintain a job that has been added to the job schedules list as a result of being either a Auto Day-Start or Auto Day-End Job. Then maintaining its details within Job Schedules

will only change the details for today's run and not alter the permanent details of the job.

Change Schedule Time Of Selected Jobs To

Use this field to change the time of jobs you have selected by entering 2 in the option field alongside the job.



Note: If you schedule many jobs to be executed at the same time then the sequence number will be used to determine the order in which the jobs should be executed.

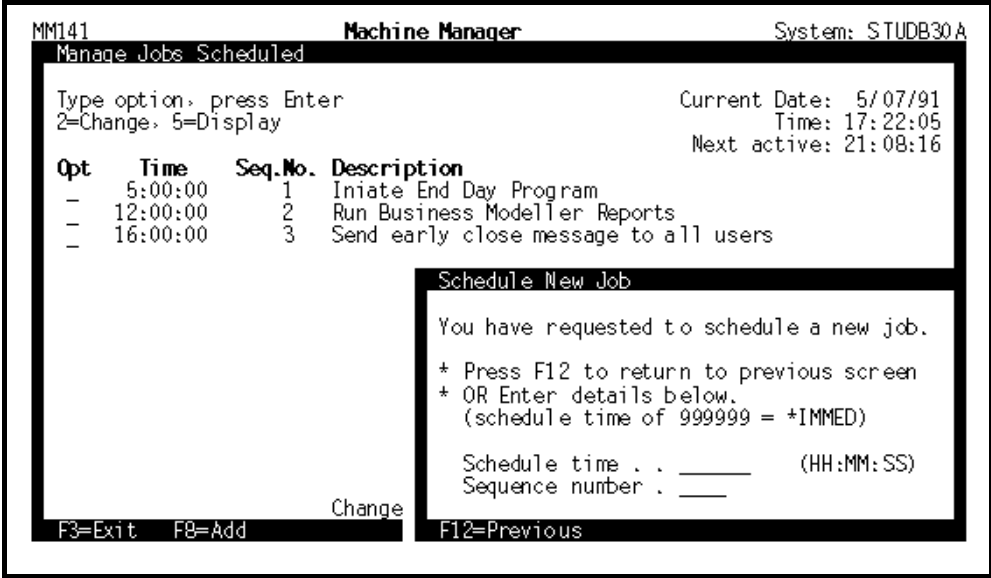


Select **F8** to create addition jobs to be run today.

Once a scheduled job has completed processing it is automatically removed from the scheduled job list.

Adding And Maintaining Scheduled Jobs (F8)

 Select **F8=Update** to add a new job or to maintain an existing job.



MM141 Machine Manager System: STUDB30A

Manage Jobs Scheduled

Type option, press Enter
2=Change, 5=Display

Current Date: 5/07/91
Time: 17:22:05
Next active: 21:08:16

Opt	Time	Seq.No.	Description
-	5:00:00	1	Initiate End Day Program
-	12:00:00	2	Run Business Modeller Reports
-	16:00:00	3	Send early close message to all users

Change

F3=Exit F8=Add F12=Previous

Schedule New Job

You have requested to schedule a new job.

* Press F12 to return to previous screen
* OR Enter details below.
(schedule time of 999999 = *IMMED)

Schedule time . . . (HH:MM:SS)
Sequence number . . .

Fields:

Scheduled Time

The time you wish the job to be executed.

Sequence Number

If you have many jobs scheduled to run at the same time then the sequence number determines the order of execution.

 Enter the time and sequence.

Manage Job Schedules Screen

Use this screen to enter the details for the job.

```

MM141                               Machine Manager                               System: STUDB30A
Manage Job Schedules
Sequence Number . . . 0004
Description . . . . A new job
Scheduled time. . . . 17:00:00      (999999=Immediately)
Request data. . . . . REQUEST DATA
                                     _____
                                     _____
                                     _____
Job name. . . . . JOBNAME           (leave blank if not a batch job)
Job user. . . . . *CURRENT          (*JOBID, *CURRENT)
Job queue . . . . . QBATCH
Job description . . QBATCH / *LIBL   (Job desc./ Library)
Job priority. . . . 5
Message queue . . . . QSYSOPR       (message queue name or *NONE)
F3=Exit  F4=Prompt  F12=Previous
  
```

Fields:

Desc

The textual description associated with the job. This is a mandatory entry.

Scheduled Time

Enter the time the job is to be executed. A time of 99:99:99 means that the job will be run immediately.

Request Data

Specify a valid command which the job is to execute.



*Note: Select **F4=Prompt** to either prompt on a command you have entered or if you have not entered a command, display the Standard IBM Major Command Groups menu so you can select any standard IBM command.*

Job Name

The Job Name if the job is to be submitted.



Tip: If a job name is entered you will be prompted to enter the following details:

Job User

The name of the user profile associated with the job when it is processed by the system.

Job Queue

The name of the job queue in which the job is placed.

Job Description

The description of this job.

Job Priority

The job queue scheduling priority. Valid values are 1 to 9, where 1 is the highest priority and 9 the lowest priority.

Message Queue

The name of the message queue to which messages concerning this job should be sent.



Press **Enter** to save the details you have entered.

End Day Times

The purpose of the End-Day Times is to control the time when the Day-End processing is initiated.

The End-Day function will at the current end day initiation time add those Auto Day-End jobs that are due to be run today and do not have an explicit schedule time to the Job Scheduler. See the Job Schedules chapter.

The Standard Job SBMENDDAY is supplied with Machine Manager to control the end day initiation. You should set this up as an Auto Day-Start Job. This job will then be added to the Job Scheduler by the Machine Manager System Start Up program as part of the Day Start Jobs and will schedule the end-day process to begin at the current end day initiation time. See the following sections in this chapter: Auto Day Start; Standard Jobs and Job Schedules.



Note: You can check that the End of Day Jobs are Scheduled by selecting option 3 Job schedules.

End Day Parameters

Select option 4 from the Main Menu and press **Enter**.

```
MMGENDDAY                      Machine Manager                      System: STUDB30A
Manage End Day Initiation

Current ENDDAY initiation time. . 19:00:00
Normal ENDDAY initiation time . . 19:00:00
Status. . . . . Waiting
Warning delay time (minutes). . . 10      (0 = No Warning)

ENDDAY job . . . . . 473671 QPGMR      JOBSCH      ACTIVE
Submitted by . . . . . 473671 QPGMR      JOBSCH      ACTIVE

Initiator started. . . . 26/04/94 20:10:15
RUNADEJOB started. . . . 26/04/94 20:13:40

F3=Exit  F5=Refresh  F6=End Day Messages  F7=Resubmit  F8=Update
F9=Execute now  F10=Release  F11=Hold  F13=Change job  F17=Workwith job
F18=End the job
```

Fields:

Current End-Day Initiation Time

The current time specified for the end-day initiation program to commence.

Normal End-Day Initiation Time

The system default time specified for the end-day initiation program to commence.



*Note that to change the time, it is necessary to change the Job Scheduler. This is done automatically by entering the new time and simply pressing **F7=Resubmit**.*

Status

The current job status. When the auto day-start job runs, the status will be shown as SUBMITTED, quickly followed by one of WAITING. When the delay time has been reached, then the status becomes RUNNING. On completion of the auto day-end jobs the status becomes NOT ACTIVE.



Note: If the status remains SUBMITTED for any length of time (more than the time taken to transfer from the job queue to the subsystem) then you should investigate the reason for this.



Warning Delay Time Specify the warning delay time (in minutes). A message will then be sent to all active screens warning of the

time remaining before end-day initiation. The status will be WARNING, when the warning time elapses the status will change to ACTIVE.

If the Day End Warning message is issued and you wish to carry on normal processing and delay the Day End then you should do one of the following:

1 Amend the current ENDDAY initiation time by using this End Day Times Function. Then use **F7=Resubmit** to re-schedule



Note: The command MNGENDDAY provides direct access to this function.

2 Use Job Scheduler and amend the scheduled time. See the Job Schedules section in this chapter.

Endday Job

The job number, user name of the user who submitted the job, and job name of the end day job.

Submitted By

The job number, user name of the user who submitted the job, and the job name of the job which submitted the end day command.

Initiator Started

The date and time when the initiator last started. This is the job which issues a warning message that the auto day-start jobs are about to commence.

Runadejob Started

The date and time which the auto day-end job last commenced.

Functions:

F6

Display messages from message queue ENDDAY, relating to this process. This details all of the messages sent as part of the End-Day job.

F7

Re-submit the job.

F9

Start the job now, regardless of the initiation time.

F10

Release the Job

F11

Hold the job

F13

Change the Job

F17

Work with the job

F18

End the Job



Note: Some functions will only be displayed if the job is active.



Press **F8=Update** to update any changes made.

Power Down Times

The purpose of the Power Down Times function is to enable the parameters which control the automatic power down of your machine to be maintained. You can change these values permanently or simply override them for the next day.

To initiate the power down of your system you can specify the command SBMPWRDWN as the request data for an Auto Day-End Job. This has been set up as a standard job. See the sections Standard Jobs and Maintaining Auto Day-End Jobs.

After the machine has powered down the machine will re-start the system immediately or at a specified time and date. When the system has re-started you will want it to run those jobs which you have set up as Auto Day-Start Jobs. To implement this automatic scheduling of jobs you need to change the system value which contains the name of the start up program to a Machine Manager program (MMSTRPGM in library IPGMMP4). To do this enter the command:



```
CHGSYSVALSYSVAL(QSTRUPPGM)VALUE(IP  
GMMP4/MMSTRPGM)
```



Note: This will permanently change the system value of the start up program.

The Power Down parameters are held in a data area. For further details see the Machine Manager - Power Down Parameters section.

Power Down Parameters



Select option 5 from the Main Menu and press **Enter**.

```
MMGFWRDWN      Machine Manager      System: STUDB30A
Manage Power Down Parameters - Normal Parameters

Power down controlled or immediate . . . *CNTRLD  *CNTRLD or *IMMED
Delay time if controlled (seconds) . . . 600      0 - 99999 or *NOLIMIT
Restart after power down . . . . . *YES        *YES or *NO

Auto-IPL (blank=none) date . . . . .          (999999=Next day)
time . . . . .

Do not power down between . . . . . 7:00:00
and . . . . . 19:30:00

Time to execute PWRDWN SYS . . . . . 5:30:00
Run RUNADJOB if powerdown not executed. *YES

Last run details:
Submitted . 24/06/91 20:07:34
Last run . 21/06/91 5:30:00
Action . . PWRDWN SYS

F3=Exit  F5=Refresh  F6=End Day Messages  F8=Update  F10=Overrides
```

Fields:

Power Down Controlled Or Immediate

*CNTRLD

All active jobs are ended in a controlled manner, based on the delay time parameter.

*IMMED

All active jobs are completed immediately.

Delay Time If Controlled

The time, in seconds, that the system will allow a controlled end to active sub-systems. If job routines are not finished in the specified delay time, any remaining jobs are ended immediately.



*Note: You can specify *NOLIMIT to ensure that all jobs are completed before the machine is powered down.*

Restart After Power Down

*NO

The system ends and powers down.

*YES

The system ends and powers down, then restarts (IPL) in unattended mode.

Auto Ipl Date/Time

Specifies the date (if required) and time for IPL. This will change the system value QIPLDATTIM.

Do Not Power Down Between/And

Specifies the time range during which the system will not power down. This would usually be during normal working hours.

Time To Execute Pwrdownsys

Specifies the time to execute the power down system command. This cannot be during the time specified above. If entered and the Day-End jobs complete before the power down time, then the job will wait until the given time before powering down.

Run Runadsjob If Powerdown Not Executed

*YES

If the power down system command is not executed, then the Auto Day-Start jobs are still run.

*NO

If the power down system command is not executed, then the Auto Day-Start jobs are not run. Note that if the time limit is exceeded, then the cycle must be re-started manually by running the Auto Day-Start jobs (see the section Maintaining Auto Day-Start Jobs).

Functions:

F6

Display messages from message queue ENDDAY, relating to this process.

F10

The override parameter fields may be used to cause the next Power Down to use different parameters. These are cleared during the Power Down sequence in order that subsequent executions of the Power Down command will use the normal parameters.



Select **F8=Update** to update any changes made.

Job Schedule Times

This function controls the delay time between which the job schedules are examined for processing. See the Job Schedules section in this chapter for more details about Job Schedules. The sleeper job performs the following function:

1. Looks at the schedule of jobs to run. See the Job Schedules section.
2. It determines if there are any jobs to be run by selecting those jobs with a scheduled time that is less than the current time and greater than the time the sleeper was last activated.
3. The jobs are then executed in time and sequence number order.
4. Once a job has been executed its details are deleted from the job schedules file.



Note: If you enter a job to be scheduled at a time that is prior to the last activation time it will be run tomorrow.



Warning: If you schedule interactive jobs, rather than batch jobs, the sleeper job is active for the whole time the interactive job is active. This could severely disrupt the expected sequencing of jobs.

Maintenance Of Job Schedule Times



Select option 6 from the Machine Manager main menu.

MNGSCHTIM		Machine Manager	System: STUDB30A
Manage Jobs Schedule Time			
Current delay time (secs)	...	300	
Normal delay time (secs)	...	300	
Status	...	Waiting	
Last activated date	...	24/06/91	
time	...	21:03:16	
Last completed date	...	24/06/91	
time	...	21:03:16	
Next activation date	...	24/06/91	
time	...	21:08:16	
F3=Exit F7=Run Day Start F8=Update			

Fields:

Current Delay Time

Use this field to change the delay time between each activation of the sleeper job. The time is expressed in seconds.

Normal Delay Time

Use this field to change the normal delay time between each activation of the sleeper job. The time is expressed in seconds.

Status

The sleeper job could be in any of the status types:

Waiting - the sleeper is inactive and waiting for its next activation.

Processing - the sleeper job has been activated and is now processing scheduled jobs.

Last Activated

The date and time the sleeper job was last activated.

Last Completed

The date and time the sleeper job last completed processing.



Note: If you are scheduling batch jobs to be submitted then this is not the time when the longest running batch job completed but the time when the sleeper job had completed the

submission of those jobs.

Next Activation

The date and time the sleeper job will next be activated.



Note: This can be altered temporarily by changing the Current delay time or permanently by changing the Normal delay time.

Functions:

F7

If for whatever reason the sleeper job has not itself been submitted then use F7 to activate the job.



Note: The sleeper job will normally be activated by the Auto Day-Start routines.



Press **F8=Update** to save the changes.

Group Schedules

The Group Schedules function enables you to create and maintain a set of dates which determine when you wish a specific job or a number of jobs to be run. The jobs are set up as either day-start or day-end jobs (see the Auto Day-Start Jobs and Auto Day-End Jobs sections of this chapter).

Maintenance Of Group Schedules



Select option 7 from the Machine Manager main menu and press **Enter**.

MM035 Machine Manager System: STUDB30 A

Maintain Group Schedule

Group Schedule Code _____

F3=Exit F4=Browse F19=Header

Fields:

Group Schedule Code

Enter the code of the group schedule you wish to maintain. If you wish to create a new group schedule enter a new four character code.

Functions:

F19

Displays a screen which enables you to maintain the 50 character description of each job schedule.



Press **Enter**. If you are creating a new group schedule, enter a 50 character description. If you are maintaining an existing schedule, enter the dates, see the Maintaining and Creating Dates section .

Maintaining And Creating Dates



Select the group schedule you wish to maintain.

MM035 Machine Manager System: STUDB30 A

Maintain Group Schedules

Group	Description	Action
INV01	Inventory Reports Schedule	*UPDATE

Type changes: press Enter

Seq.No.	Date	Seq.No.	Date	Seq.No.	Date
10	_____	110	_____	210	_____
20	_____	120	_____	220	_____
30	_____	130	_____	230	_____
40	_____	140	_____	240	_____
50	_____	150	_____	250	_____
60	_____	160	_____	260	_____
70	_____	170	_____	270	_____
80	_____	180	_____	280	_____
90	_____	190	_____	290	_____
100	_____	200	_____	300	_____

F3=Exit F9=Generate dates F12=Previous

Fields:

Date

You may enter a maximum of 60 dates against sequence numbers. The date entered must be valid.



*Note: If you enter dates out of sequence they will automatically be re-sequenced for you. Therefore if you wish to add a date into the middle of an existing sequence add the date to the end of the list and then press **Enter** to re-sequence.*



Press **F9=Generate Dates**. The generate dates facility is provided to automatically generate a set of dates based on given parameters.

Generate Dates



Select **F9=Generate Dates** to display the Generate Dates screen.

MM035 Machine Manager System: STUDB30A

Maintain Group Schedules

Group	Description	Inventory Reports	Schedule	Action
INV01				*UPDATE

Type changes, press Enter

Generate dates

Start date for the sequence . . . 50791

Number of dates to generate . . .

Specify ONE of the following:-

 Last day in each month . . .

 A day in each month . . .

 Number of days interval . . .

F3=Exit F12=Previous

Seq. No.	Date	Se
10		
20		
30		
40		
50		
60		
70		
80		
90		190
100		200

F3=Exit F9=Generate dates F12=Previous

Fields:

Start Date For The Sequence

Enter the date to start the sequence of generated dates from. This field is defaulted to today's date.

Number Of Dates To Generate

Enter the number of days, between 1 and 60, to automatically generate.

Last Day In Each Month

Enter 1 if you wish the dates generated to be the last day of each month.

A Day In Each Month

To run jobs on a specific day number in each month then enter that day number.



Note: The day numbers 29, 30, and 31 could be invalid depending how many dates you generate and where you start the generation.

Number Of Days Interval

To generate a series of dates which have a set interval between them enter the number of days. The interval entered may be between 1 and 99 days.



Press **Enter** to generate the dates.



Note: A set of automatically generated dates can be maintained manually or re-generated.

Protected Queues

All output, job or message queues except those on a protected list, may be cleared by use of Auto Day-End standard jobs. See the sections Standard Jobs and Maintaining Auto Day-End Jobs.

Any job, message, or output queue may be protected from clearance.



Select option 8 from the Main Menu and press **Enter** to display the Protected Queues options.

Protected Job Queues



Select option 1 within Protected Queues and press **Enter**.

MM041 Machine Manager System: STUDB30A

Manage Protected Job Queues

Enter names and libraries of queues to be protected.
If a day is set to 1 then the queue will be cleared on that day.
If a day is set to 0 then the queue will not be cleared on that day.

Queue Library	Queue Name	Day Mask
QGPL	QBASE	0000000
QGPL	QBATCH	0000000
QGPL	QPMR	0000000

F3=Exit Bottom

Fields:

Queue Library

Enter the name of the library containing the queue to be maintained.

Queue Name

Enter the name of the queue to be maintained.

Day Mask

A day mask indicating on which day(s) you wish the job to be executed. A 1 indicates that the job should be performed and a 0 indicated that it should not.

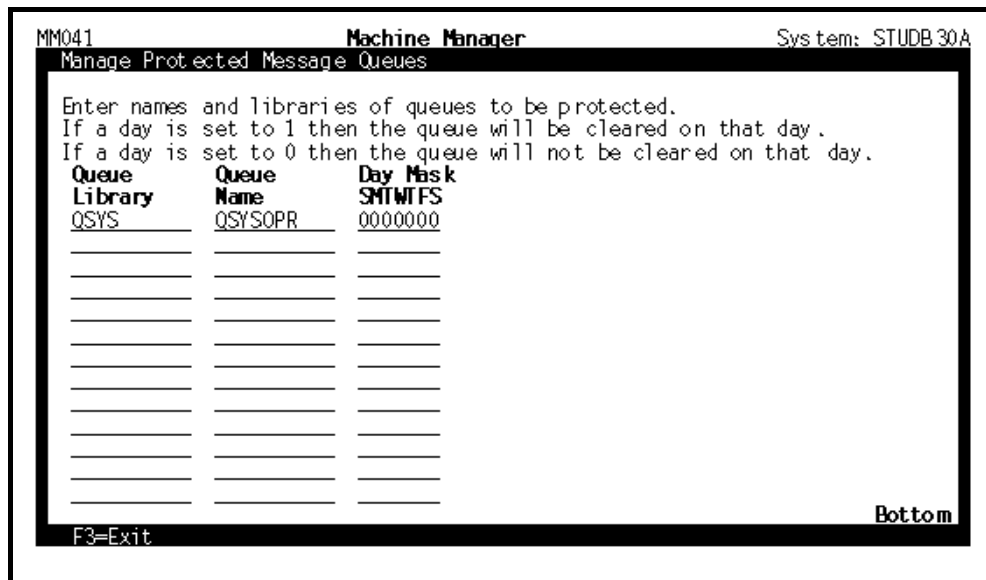


Press **Enter**.

Any job queues not included on this list will be cleared automatically if you have included the standard job Clear unprotected job queues as an Auto Day-End job. See the sections Standard Jobs and Maintaining Auto Day-End Jobs.

Protected Message Queues

 Select option 2 within Protected Queues and press **Enter**.



MM041 Machine Manager System: STUDB 30A

Manage Protected Message Queues

Enter names and libraries of queues to be protected.
If a day is set to 1 then the queue will be cleared on that day.
If a day is set to 0 then the queue will not be cleared on that day.

Queue Library	Queue Name	Day Mask SMTWTFSS
QSYS	QSYSOPR	0000000

F3=Exit Bottom

Fields:

Queue Library

Enter the name of the library containing the queue to be maintained.

Queue Name

Enter the name of the queue to be maintained.

Day Mask

A day mask indicating on which day(s) you wish the job to be executed. A 1 indicates that the job should be performed and a 0 indicated that it should not.

 Press **Enter**.

Any job queues not included on this list will be cleared automatically if you have included the standard job Clear unprotected message queues as an Auto Day-End job. See the sections Standard Jobs and Maintaining Auto Day-End Jobs.

Protected Output Queues



Select option 3 within Protected Queues and press **Enter**.

MM041 Machine Manager System: STUDB30A

Manage Protected Output Queues

Enter names and libraries of queues to be protected.
If a day is set to 1 then the queue will be cleared on that day.
If a day is set to 0 then the queue will not be cleared on that day.

Queue Library	Queue Name	Day Mask
IPGSYS	SAVPRT	0000000
QGPL	QPRINT	0000001

F3=Exit Bottom

Fields:

Queue Library

Enter the name of the library containing the queue to be maintained.

Queue Name

Enter the name of the queue to be maintained.

Day Mask

A day mask indicating on which day(s) you wish the job to be executed. 1 indicates that the job should be performed and 0 indicated that it should not.



Press **Enter**.

Any job queues not included on this list will be cleared automatically if you have included the standard job Clear unprotected output queues as an Auto Day-End job. See the sections Standard Jobs and Maintaining Auto Day-End Jobs.